FIBRE PRODUCTS OF REPRESENTABLE FUNCTORS WITH A PROPERTY MAKE THE DIAGONAL REPRESENTABLE WITH THAT PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains one xymatrix display with two cartesian squares; it is part of the counted proof span and is preserved verbatim, with the chapter's own Xy-pic setup retained in the preamble.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Representability of the diagonal tested on pairs of maps. Lemma lemma-representable-diagonal (921-934), used at proof lines 973-974. Gives representability of Delta.
- Morphisms of schemes correspond to representable transformations with the same property. Lemma lemma-morphism-schemes-gives-representable-transformation-property, used at proof lines 975-978. Reformulates condition (2) as a statement about transformations.
- Stability of the property under base change. Definition definition-relative-representable-property, used at proof lines 990-994. Transfers property P to the base change of Delta.

Source reference. Original definition 025V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2828–2828. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3806–3806. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 025W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2829–2829. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let F be a presheaf of sets on $(Sch/S)_{fppf}$. Let $\mathcal{P}$ be a property as in Definition 025V. If for every $U, V \in \mathrm{Ob}((Sch/S)_{fppf})$ and $a \in F(U)$, $b \in F(V)$ we have*

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) $h_U \times_{a,F,b} h_V$ *is representable, say by the scheme* W*, and*
(2) *the morphism* $W \to U \times_S V$ *corresponding to* $h_U \times_{a,F,b} h_V \to h_U \times h_V$ *has property* $\mathcal{P}$*,*

then $\Delta : F \to F \times F$ *is representable and has property* $\mathcal{P}$*.*

Proof. Observe that Δ is representable by Lemma 025W. We can formulate condition (2) as saying that the transformation $h_U \times_{a,F,b} h_V \to h_{U \times_S V}$ has property $\mathcal{P}$, see Lemma 02WJ. Consider $T \in \mathrm{Ob}((Sch/S)_{fppf})$ and $(a, b) \in (F \times F)(T)$. Observe that we have the commutative diagram

$$\begin{array}{ccc}
F \times_{\Delta, F\times F, (a,b)} h_T & \longrightarrow & h_T \\
\downarrow & & \downarrow{\scriptstyle \Delta_{T/S}} \\
h_T \times_{a,F,b} h_T & \longrightarrow & h_{T \times_S T} \\
\downarrow & & \downarrow{\scriptstyle (a,b)} \\
F & \xrightarrow{\Delta} & F \times F
\end{array}$$

both of whose squares are cartesian. In this way we see that the morphism $F \times_{F\times F} h_T \to h_T$ is the base change of a morphism having property $\mathcal{P}$ by $\Delta_{T/S}$. Since $\mathcal{P}$ is preserved under base change this finishes the proof. □